Department of Mathematics, IIT Madras

MA 2102 - Differential Equations, ODD Semester, 2026-2027

Assignment-6 : Bessel Functions

---

1. Show that for integer values of n, $J_{-n}(x) = (-1)^n J_n(x)$ and $J_n(-x) = (-1)^n J_n(x)$.

2. Prove the following results:

   (a) $J_{3/2}(x) = \sqrt{\dfrac{2}{\pi x}}\left[\dfrac{\sin x}{x} - \cos x\right]$; (b) $J_{3/2}(x) = x^{-1}J_{1/2}(x) - J_{-1/2}(x)$;

   (c) $J_2'(x) = \dfrac{1}{2}\big[J_1(x) - J_3(x)\big]$; (d) $J_3 + 3J_0' + 4J_0''' = 0$;

   (e) $J_0''(x) = \dfrac{1}{2}\big(J_2(x) - J_0(x)\big)$; (f) $J_1''(x) = -J_1(x) + \dfrac{1}{x}J_2(x)$;

   (g) $J_{1/2}(x)\,J_{-1/2}(x) = \dfrac{\sin 2x}{\pi x}$; (h) $J_{1/2}(x)^2 + J_{-1/2}(x)^2 = \dfrac{2}{\pi x}$.

3. Show that

   (a) $\displaystyle\int x^4 J_0(x)\,dx = (x^4 - 9x^2)J_1(x) + (3x^3 - 9x)J_0(x) + 9\int J_0(x)\,dx$;

   (b) $\displaystyle\int J_5(x)\,dx = -2J_4(x) - 2J_2(x) - J_0(x) + C$.

   (c) $\displaystyle\int x^{-2}J_5(x)\,dx = -x^{-2}J_4(x) + 2x^{-3}J_3(x) + C$;

   (d) $\displaystyle\int x^p J_\nu\,dx = x^p J_{\nu+1}(x) - (p - \nu - 1)\int x^{p-1}J_{\nu+1}(x)\,dx$.

4. If y_1 and y_2 are any two solutions of Bessel equation of order ν, then prove that $y_1y_2' - y_1'y_2 = \dfrac{c}{x}$ for a suitable constant c.

5. Prove that between each pair of consecutive positive zeros of $J_\nu(x)$, there is exactly one zero of $J_{\nu+1}(x)$, and one zero of $J_{\nu-1}(x)$.

6. Solve each of the following, using the indicated transformation, in terms Bessel functions:

   (a) $y'' + 4xy = 0$, $\left(y = u\sqrt{x},\ \dfrac{4}{3}x^{3/2} = z\right)$.

   (b) $x^2y'' - 3xy' + 4(x^4 - 3)y = 0$, $(y = x^2u,\ x^2 = z)$.

   (c) $x^2y'' + xy' + (\lambda^2x^2 - \nu^2)y = 0$, $(\lambda x = z)$.

   (d) $y'' + e^{-x}y = 0$, $\left(z = 2e^{-\frac{x}{2}}\right)$.

   (e) $xy'' + (1 + 2p)y' + xy = 0$, $(y = x^{-p}u)$.

   (f) $xy'' - 5y' + xy = 0$, $(y = x^3u)$.

   (g) $xy'' - (1 + 2p)y' + xy = 0$, (Find the transformation!)

   (h) $y'' + xy = 0$ (Find the transformation!)

7. Using the recurrence relations, prove that

   (i) $\dfrac{d}{dx}\left[J_n^2(x)\right] = \dfrac{x}{2n}\left[J_{n-1}^2(x) - J_{n+1}^2(x)\right]$

   (ii) $\dfrac{d}{dx}\left[J_n^2(x) + J_{n+1}^2(x)\right] = 2\left[\dfrac{n}{x}J_n^2(x) - \dfrac{n+1}{x}J_{n+1}^2(x)\right]$

8. Express $J_4(x)$ in terms of $J_0(x)$ and $J_1(x)$.

9. Show that $\displaystyle\int_0^r xJ_0(ax)\,dx = \frac{r}{a}J_1(ar)$.

10. Using the generating function for Bessel's function, prove that $J_0^2 + 2\displaystyle\sum_{n=1}^{\infty} J_n^2 = 1$. Hence, deduce that $|J_0| \le 1$ and $|J_n| \le \sqrt{\dfrac{1}{2}}$.

11. Find the Fourier-Bessel series for the following functions over the given interval and in terms of the Bessel functions of prescribed order:

    (a) $f(x) = x$ over $[0, 1]$, $\nu = 2$. (b) $f(x) = 1$ over $[0, 3]$, $\nu = 0$.

    (c) $f(x) = x^3$ over $[0, 3]$, $\nu = 1$. (d) $f(x) = x^2$ over $[0, 2]$, $\nu = 2$.

    (e) $f(x) = \sqrt{x}$ over $[0, 1]$, $\nu = \dfrac{1}{2}$. Determine the exact roots of Bessel function in this case, and hence compute explicitly the coefficients in the Bessel series expansion.

    (f) $f(x) = \begin{cases} 1, & 0 < x < k, \\ 0, & k < x < 1; \end{cases}$ $\nu = 0$. Discuss the convergence of the series in the interval $0 < x < 1$.

12. Let a, b, c, and ν be positive reals such that $a \neq b$. Then prove that

$$\int_0^c J_\nu(ax)J_\nu(bx)\,x\,dx = \frac{c}{b^2 - a^2}\left[aJ_\nu(bc)J_{\nu-1}(ac) - bJ_\nu(ac)J_{\nu-1}(bc)\right].$$

13.* Compute the explicit spherical Bessel functions $j_0(x)$, $j_1(x)$, $j_2(x)$.

14.* Prove the following for spherical Bessel functions:

    (a) $\dfrac{d}{dx}\left[x^{n+1}j_n(x)\right] = x^{n+1}j_{n-1}(x)$; (b) $\dfrac{d}{dx}\left[x^{-n}j_n(x)\right] = -x^{-n}j_{n+1}(x)$;

    (c) $j_{n-1}(x) + j_{n+1}(x) = \dfrac{2n+1}{x}j_n(x)$; (d) $nj_{n-1}(x) - (n+1)j_{n+1}(x) = (2n+1)j_n'(x)$.

13. From Jacobi series, prove that (a) $\sin x = 2J_1(x) - 2J_3(x) + 2J_5(x) - \ldots$

    (b) $\cos x = J_0(x) - 2J_2(x) + 2J_4(x) - 2J_6(x) + \ldots$ (c) $1 = J_0 + 2J_2 + 2J_4 + 2J_6 + \ldots$

14. Using integral representation of Bessel's function, compute $\displaystyle\int_0^\infty e^{-ax}J_0(bx)\,dx$.